# Predictive Modeling for Heart Disease Diagnosis Using Machine Learning Algorithms


## ABSTRACT

Heart disease remains a leading global cause of death, necessitating rapid and accurate diagnostic support tools. This research implements four machine learning modelsLogistic Regression, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) to predict heart disease using 14 clinical features.

From 1,025 collected clinical records, 723 duplicates were identified and removed, resulting in 302 unique patient records used for modeling. After preprocessing and feature scaling, models were trained using an 80:20 split.

Logistic Regression emerged as the best-performing model, achieving:

- Accuracy: 0.8033
- Precision: 0.8000
- Recall: 0.8485
- F1-score: 0.8235

ROC-AUC analysis further confirmed strong discriminative ability (AUC = 0.8712). Feature importance ranking (Random Forest) identified chest pain type (cp), max heart rate (thalach), and major vessels (ca) as the strongest predictors.

The results demonstrate that interpretable ML models particularly Logistic Regression remain highly effective for structured clinical datasets.


## I. INTRODUCTION

### A. Background

Heart disease continues to produce significant global mortality and economic burden. Despite widespread diagnosis tools, early detection remains challenging and resource-intensive.

### B. Objectives

1. Compare multiple ML algorithms for heart disease prediction
2. Evaluate performance with standardized metrics
3. Identify the most influential predictors
4. Provide a reproducible clinical ML workflow

## II. DATASET SUMMARY

### A. Dataset Characteristics

- Initial records: 1,025
- Columns: 14
- Duplicates: 723 removed → 302 unique rows
- Training set: 241
- Testing set: 61
- Missing values: None

## B. Features

| Variable | Description |
|---|---|
| age | Age in years |
| sex | 1 = male, 0 = female |
| cp | Chest pain type |
| trestbps | Resting blood pressure |
| chol | Serum cholesterol |
| fbs | Fasting blood sugar |
| restecg | ECG results |
| thalach | Maximum heart rate |
| exang | Exercise-induced angina |
| oldpeak | ST depression |
| slope | ST segment slope |
| ca | Number of vessels |

| | |
|---|---|
| thal | Thalassemia |
| target | 1 = Disease, 0 = No disease |

# III. DATA PREPROCESSING

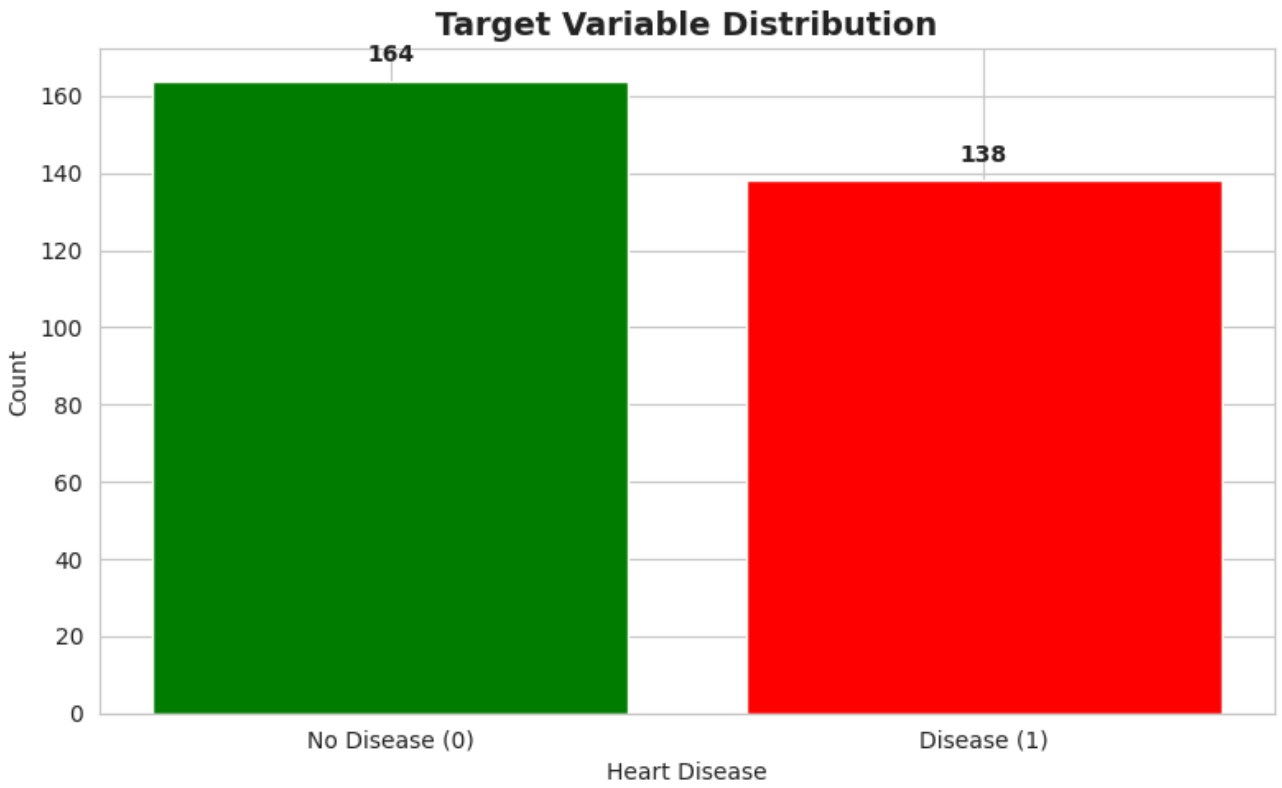
- Duplicate removal drastically reduced dataset size
- All fields contained valid numeric values
- StandardScaler applied to all features
- Train-test split used stratified sampling

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# IV. EXPLORATORY DATA ANALYSIS

## A. Target Distribution

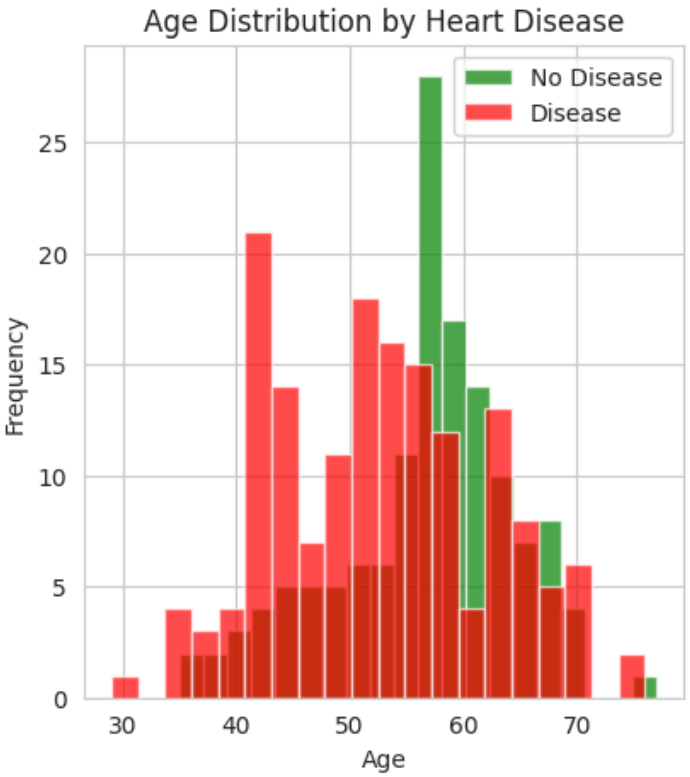
- Disease (1): 138
- No Disease (0): 164



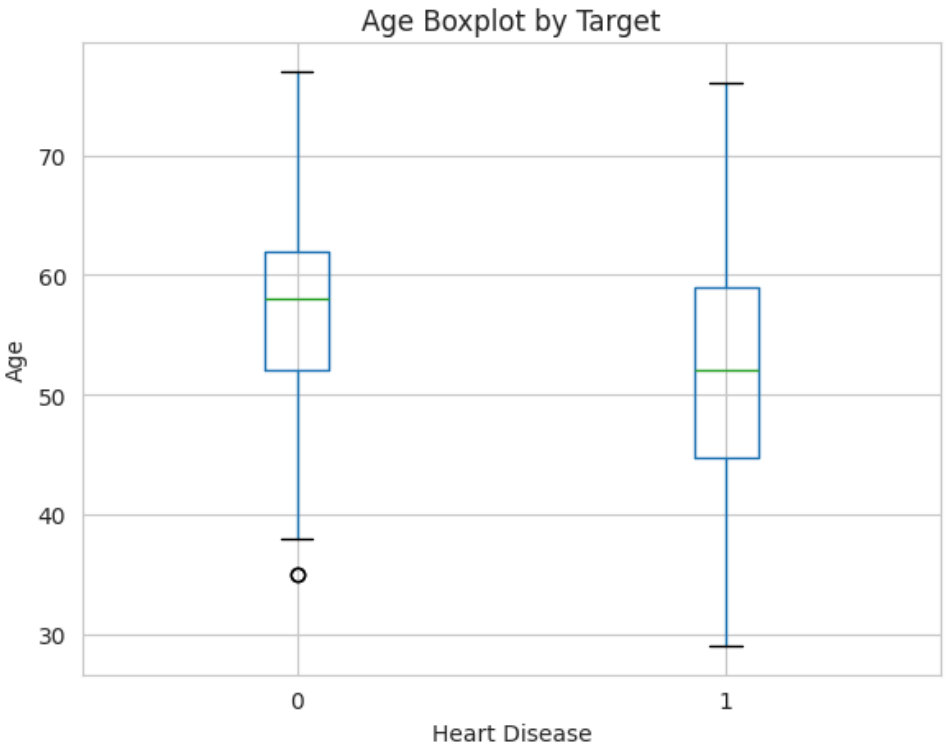

*Target Variable Distribution*

# B. Age Distribution Insights

- Patients with heart disease tend to be slightly older
- Median age higher for target=1 group




*Histogram age by target*

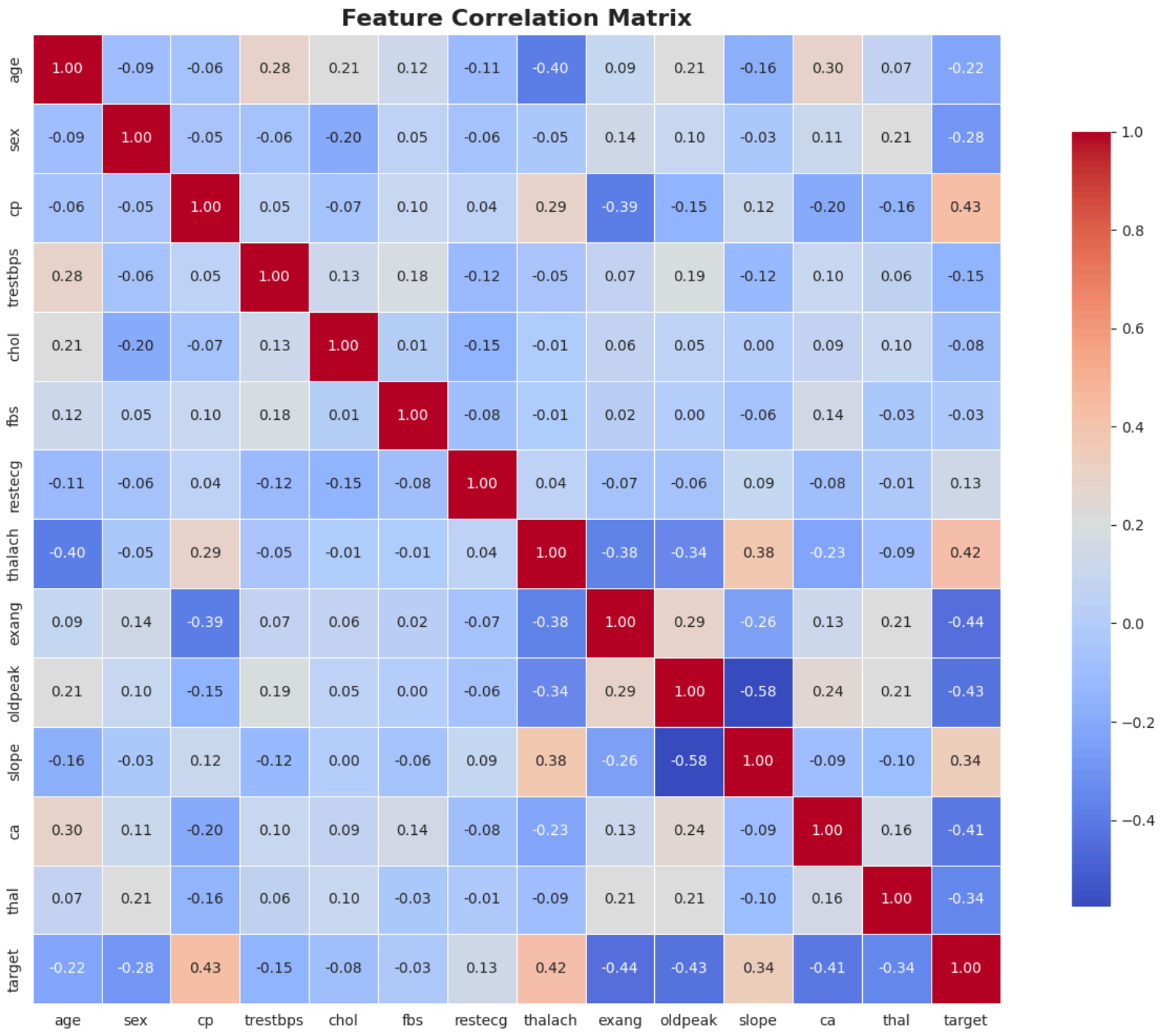



*Boxplot age by target*

# C. Correlation Matrix

Key observed correlations:

- thalach negatively correlated with disease
- cp, oldpeak, exang, ca positively correlated with disease




*Feature Correlation Matrix Heatmap*

# V. MODEL DEVELOPMENT

## A. Models Trained

- Logistic Regression
- Random Forest
- Support Vector Machine
- K-Nearest Neighbors

## B. Evaluation Metrics

All models evaluated on:
Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix

# VI. RESULTS

## A. Performance Summary

| Model | Accuracy | Precision | Recall | F1-score |
|---|---|---|---|---|
| Logistic Regression | 0.8033 | 0.8000 | 0.8485 | 0.8235 |
| Random Forest | 0.7541 | 0.7647 | 0.7879 | 0.7761 |
| SVM | 0.7705 | 0.7714 | 0.8182 | 0.7941 |
| KNN | 0.7869 | 0.7778 | 0.8485 | 0.8116 |

| | | | | |
|---|---|---|---|---|
| Random Forest (Tuned) | 0.7869 | 0.7941 | 0.8182 | 0.8060 |

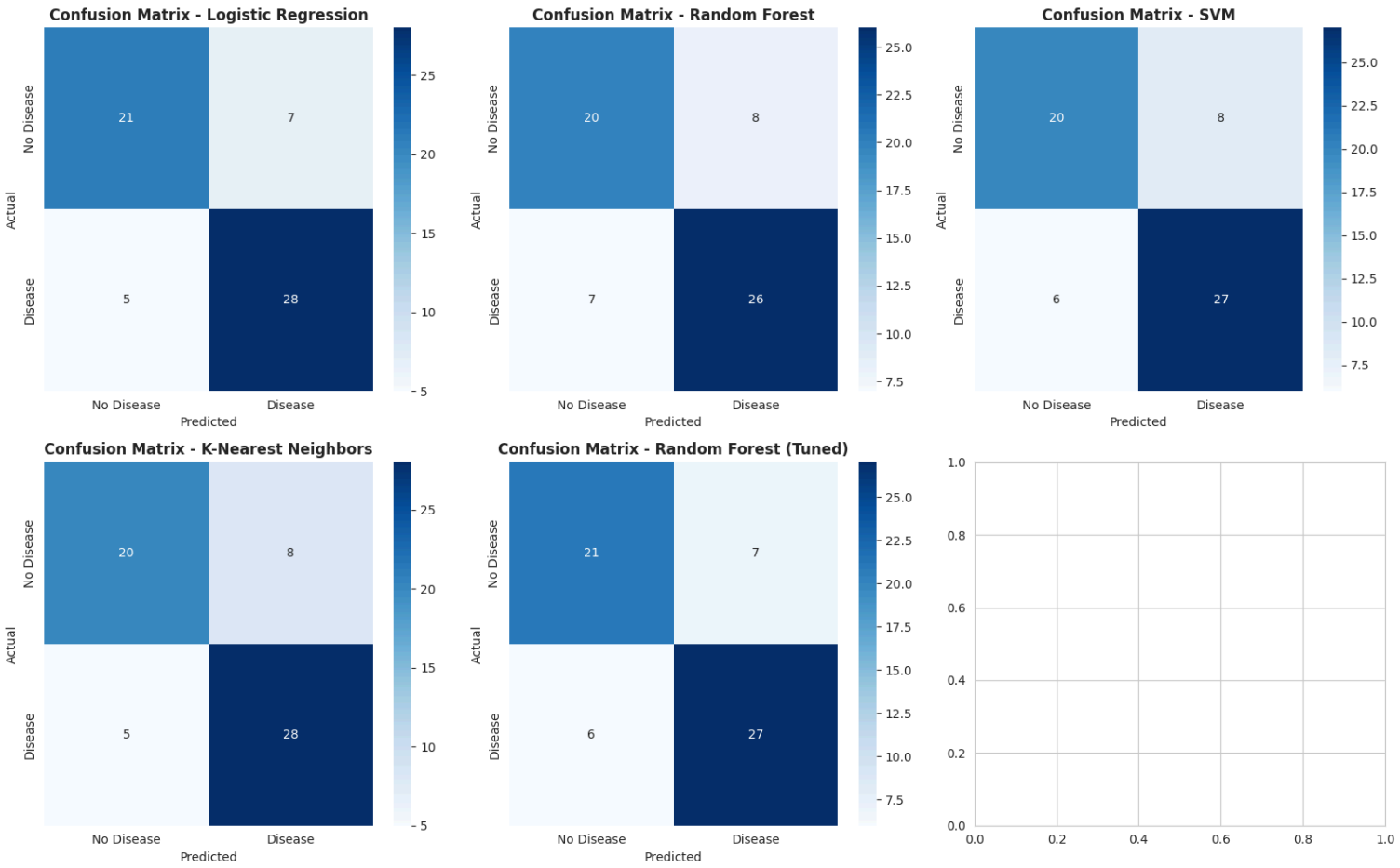
Best Model: Logistic Regression (Highest accuracy & F1)

## B. ROC-AUC

Actual results (page 9):

- Logistic Regression: 0.8712
- Random Forest: 0.8588
- SVM: 0.8420
- KNN: 0.8377
- RF Tuned: 0.8636

## C. Confusion Matrices

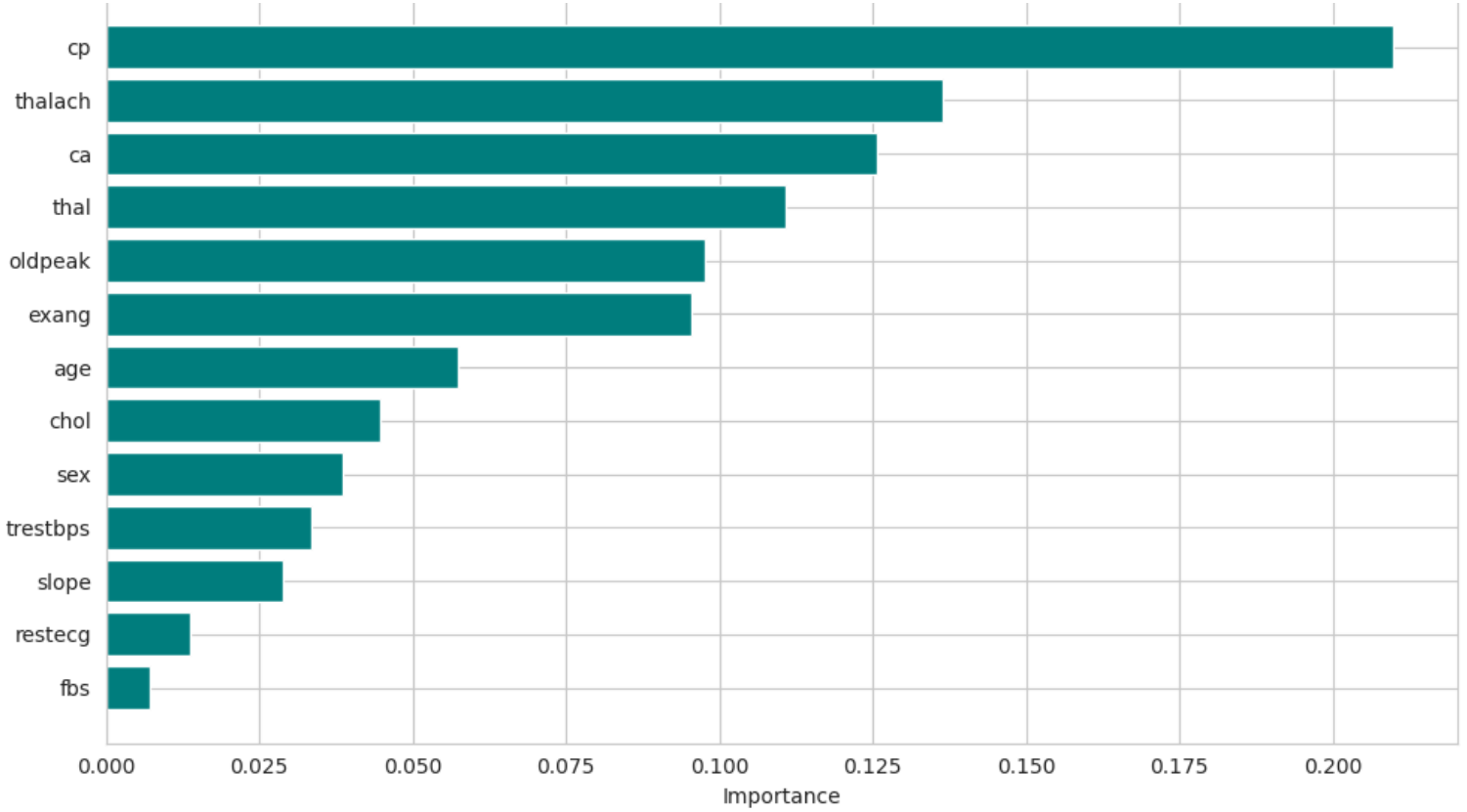



*Confusion matrices grid for all models*

## D. Feature Importance (Random Forest)

Top features:

1. cp
2. thalach
3. ca
4. thal
5. oldpeak
6. exang




*Feature importance - Tuned Random Forest*

# VII. DISCUSSION

- Logistic Regression performed best, likely due to linear separability in dataset
- Exercise-related features (thalach, exang, oldpeak) show strong predictive power
- Duplicate-heavy dataset required major cleaning
- Random Forest tuning improved performance but still below Logistic Regression

# VIII. CONCLUSION

- Logistic Regression is the best-performing model
- Strong AUC (0.8712) indicates reliable discrimination
- Key clinical predictors align with medical expectations
- The pipeline demonstrates a robust and reproducible workflow

# REFERENCES


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